Radar gestures, start to submission

Classifying four hand gestures from micro-Doppler spectrograms — the data, the decisions that carry the marks, and how two people build one report together.

Samples	People	Classes	Report cap	Due
960	6 (A–F)	4	10 pages	Week 8

1. What is being asked

Build a pipeline that names a gesture — *wave, pinch, swipe,* or *click* — **regardless of who performed it**. That clause sets the evaluation protocol (see §4). Task 1 is pre-processing (clean, normalise, optionally extract features); Task 2 is modelling, with written arguments for both the model and the validation procedure.

Admissible or fail, independent of score: unclear English, no deep neural network tested, illegible figures, or a report that skips the prescribed structure. Once admissible, neither polished English nor clean code earns **extra** points — so clear the bar on both, then spend the remaining time on argumentation and validation design.

Deliverables: a 10-page PDF report + Python code, one submission per pair, to Brightspace. Week 7 has a lecturer feedback slot — arrive with numbers.

2. What is in the data

Six archives (one per person), four gesture folders each, 40 CSVs per folder — 960 samples total, 240 per class, perfectly balanced. Label is in the filename: `PersonAg4s1.csv` = person A, gesture 4 (`g1`=wave, `g2`=pinch, `g3`=swipe, `g4`=click), sample 1.

Property	Measured	Implication
Doppler axis	800 bins, fixed	No alignment needed across files.
Time axis	72–411 frames (med. 156)	Ragged — must be made uniform before batching.
Values	dB, max = 0.0, floor ≈ −41	Already per-sample normalised; only the floor is a free choice.
Energy band	bins ≈ 330–470	≈ 82% of every input is noise floor, carrying no class signal.
Sensor	Ancortek 24 GHz FMCW, 750 MHz BW	One sentence in the report shows you read the dataset docs.

Table 1: Measured on the local DopNet subset — averaged over 60 sampled spectrograms across all six people.

> OPEN QUESTION FOR WEEK 7
> The brief mentions a test set "already separated" from training; full public DopNet has 3,052 files vs. 960 here. Design validation (§4) to work entirely within the 960 you have — treat an official test set as a bonus, not a rewrite.

3. Task 1 — four pre-processing decisions

Each should be written up as *options* → *choice* → *evidence* — that structure is effectively the rubric.

D1 **Crop the Doppler axis to bins ≈ 330–470**
Outside this band the mean is within 1 dB of the noise floor (table above). Cuts input rows 800→140, shrinking the first layer 5–6× and the overfit surface with it. Verify per-sample retained energy ≥ 99%, not just on the average.

D2 **Fix the ragged time axis — resize or pad, test both**
Resize to 128 frames normalises gesture duration away (model sees shape only). **Pad/crop** with the noise floor preserves duration as a cue. These are different physical assumptions, not a style choice — run both through identical folds and report the gap.

D3 **Clip the dynamic range, fit nothing on test folds**
Clip below ≈ −40 dB (just under the noise floor), rescale to [0,1]. Sweep 2–3 clip values and state the choice. Any fitted scaler (`StandardScaler`, PCA) is fit on training folds only — fitting on everything is the classic leak.

D4 **Explicit features: take centroid + bandwidth (see §3 below)**
Optional, but a feature baseline makes Task 2′s CNN comparison a real result instead of a lone number. Of the three papers in the brief, centroid/bandwidth is the cheapest to implement and interpret — see the next section.

---

4. The three reference papers

The brief names three papers as inspiration for Task 1′s optional feature extraction. One family is enough — pick, don't implement all three.

1 — Amin, Zeng & Shan (2019), envelope features RadarConf 2019

Tracks the upper/lower **envelope** of the micro-Doppler spectrogram (the outline traced by the strongest Doppler return over time) as a compact signature, classified with **k-NN under Manhattan (L1) distance** — chosen specifically because L1 doesn't wash out small-but-informative envelope values the way L2/PCA-based distances can. Shown to beat PCA and sparse Fourier-dictionary reconstruction as baselines. **Use:** cheapest of the three to extract (two 1-D curves per sample) and a fast sanity-check baseline before investing in centroid/Chebyshev features.

---

2 — Fioranelli, Ritchie & Griffiths (2016), centroid & bandwidth moments IET RSN 10(9):1702–1710, §3.2

At each time slice of the spectrogram, computes the power-weighted **centroid** (mean Doppler bin) and the **bandwidth** (spread around it) — two time series per sample — then takes their statistical **moments** (mean, variance, skewness, kurtosis) as a fixed-length feature vector. Originally built for armed/unarmed personnel classification from multistatic radar, not gestures, but the feature definition is domain-agnostic. **Use:** the recommended family (D4) — 20 lines of NumPy, physically interpretable, good report material (a centroid-vs-time plot per gesture class is an easy, legible figure).

---

3 — Pallotta et al. (2021), Chebyshev moments MetroAeroSpace 2021, pp. 182–187

Computes **Chebyshev moments** — discrete orthogonal moments, so no continuous-approximation error, with a symmetry property that keeps computation cheap — directly on the cadence-velocity diagram (CVD) of each signal, treating it as a 2-D image rather than extracting 1-D curves first. Notably, it is tested on **this same DopNet dataset**. **Use:** the strongest match to "spectrogram as an image" framing and the closest prior work to this exact assignment — worth citing even if you don't implement it, and a good stretch goal if D4′s baseline undersells the physics features.

---

5. Task 2 — a model ladder

Model	Input	Question it answers
SVM (RBF) / kNN	Centroid-bandwidth or envelope features	Floor set by physics features alone.
MLP	Flattened, downsampled image	Does convolution earn its keep over a plain dense net?
Small CNN (required)	140×128 normalised image	Does learned spatial structure beat hand-crafted features?

Table 2: All three evaluated on identical folds — otherwise the comparison is meaningless.

With only 960 samples, keep the CNN small: 3–4 conv blocks (16/32/64 channels), batch-norm, light dropout, global average pooling, softmax — not a from-scratch ResNet. Augment with time shifts/stretches, noise, SpecAugment-style masking.

> DO NOT FLIP THE DOPPLER AXIS
> The axis is **signed velocity** — flipping reverses motion direction (a swipe-left becomes swipe-right) and corrupts the label. Standard image augmentation reflexes don't apply here; say so in the report.

Report accuracy and macro-F1 as mean ± std across folds, plus a confusion matrix (expect pinch/click to confuse each other — both short, low-velocity).

> ENVIRONMENT
> TensorFlow 2.18 is installed, no PyTorch — use Keras unless you both install Torch. Pin `numpy<2.3` in `requirements.txt`; the installed 2.5.3 already warns on `import sklearn`.

6. Validation — decides the grade

"Regardless of the person" specifies the **test distribution**: new user, not seen in training. A random 80/20 split fails this badly here, because the 40 samples per person/gesture are segments cut from one continuous 30 s recording — near-duplicates that leak across a random split and let the model recognise the **person** instead of the **gesture**.

Protocol: hold out one person entirely as a final test set → leave-one- person-out over the remaining five for model selection (5 folds, mean ± std) → one final, single run on the held-out person.

> CHEAP MARKS
> Report **both** random-split and leave-one-person-out accuracy and explain the gap — it's a real finding about inter-person variability, and costs one extra run. Expect random-split to look better and mean less.

7. Writing the report

Structure is prescribed; deviating from it is an admissibility failure.

Section	Budget	Must contain
Members / Summary	½ pg / <200 w	Names, emails, numbers; summary written last.
ML pipeline	1.5 pg	**Workflow figure** — required by the brief.
Task 1	3 pg	Four decisions as options→choice→evidence; Doppler-profile and feature-separation plots.
Task 2	4 pg	Model ladder, validation protocol + why, results ± folds, confusion matrix, random-vs-LOPO gap.
Conclusions	<200 w	What worked; what more data/time would change.

Figures: vector PDF not screenshots, ≥9pt axis labels, dB colour bars, named/unit'd axes. Check legibility in greyscale. Five careful figures beat nine cramped ones.

8. Working as a pair

The brief requires joint decisions + splittable tasks, and **either partner must be able to argue any decision** — the tooling below makes that review step cheap enough to actually happen.

> BEFORE THE FIRST COMMIT
> `C:\Users\hawma` itself is a git repo pointing at an unrelated course remote. Create a fresh repo **inside** the WICOS folder (or move the project out of the home tree) before running any git command here.

Repo: one private GitHub repo for code + report, both as collaborators. Gitignore `*.zip`, `data/`, checkpoints, `.ipynb_checkpoints` — archives stay in OneDrive/SURFdrive, not git. Pin `requirements.txt` (incl. `numpy<2.3`). Work on short feature branches, merge via PRs the other partner reviews.

Notebooks diverge badly in merges (base64 output blobs). Put real logic in plain `.py` modules (`data.py`, `features.py`, `models.py`, `evaluate.py`); keep notebooks thin — import + plot only. Use `nbstripout` if you must share notebooks directly.

> REPORT TOOLING: TYPST, IN THE REPO, VIA TINYMIST
> Lives beside the code, diffs as text, regenerated figures land on next compile. If a partner is allergic to git, use Overleaf instead rather than fighting it — don't split a Word doc for a 10-page figure-heavy report.

Split by pipeline stage, not "code vs. writing" — that split leaves one partner unable to defend the model.

Owner	Builds	Writes
A	Loader, Doppler crop, time-axis test, normalisation, feature extraction, SVM/kNN	Task 1
B	CNN, augmentation, training loop, MLP control, confusion matrices	Task 2
Both	Fold harness — build together, it decides every number	Pipeline figure, summary, conclusions

Keep `DECISIONS.md`: one line per choice, reason + evidence, appended as you go — it becomes the Week 8 argumentation sections almost verbatim.

9. Timeline

Week	Build	Milestone
5	Loader, EDA, crop band fixed from measurement	D1–D4 settled
6	Baselines, first CNN, LOPO fold harness	Fold harness frozen — don't change it after
7 (checkpoint)	Final model, ablations	Lecturer feedback — bring numbers
8	Report writing, figure polish	Submit PDF + code

First session: move the repo out of the home git tree · create GitHub repo with `.gitignore` and pinned `requirements.txt` · write `data.py` and verify 960/240-per-class · plot one spectrogram per gesture with a dB colour bar · reproduce the Doppler-profile measurement on all 960 files · write the LOPO fold generator before any model exists · start `DECISIONS.md`.
